

Question:

A 34-year-old woman presents to her OB/GYN with complaints of missing her last 3 periods as well as intermittent spontaneous milky-white nipple discharge bilaterally for the past 3 months. Vital signs are stable and within normal limits. Neurologic examination is without abnormality, including normal visual fields. Serology and MRI of the brain are ordered, with results pending. Which of the following sets of laboratory results would be expected in this patient?

Answer Choices:

A. Decreased prolactin, decreased FSH, decreased LH

B. Decreased prolactin, increased FSH, increased LH

C. Increased prolactin, decreased FSH, increased LH

D. Increased prolactin, decreased FSH, decreased LH

Reasoning + Answer:

<think>

Finding Reasoning Paths :

1 . **Galactorrhea â Prolactin-secreting pituitary adenoma (prolactinoma) â Decreased FSH/LH â Elevated prolactin levels . **

2 . **Absent menses â Abnormal uterine bleeding (likely due to hyperprolactinemia) â Decreased FSH/LH â Elevated prolactin levels . **

3 . **Spontaneous nipple discharge â Prolactin-secreting pituitary adenoma â Decreased FSH/LH â Elevated prolactin levels . **

4 . **Elevated prolactin â Hypothalamic dysfunction (e.g., empty sella) â Decreased FSH/LH . **

5 . **Normal pituitary on MRI â Lack of structural pathology â Hyperprolactinemia due to functional causes (e.g., hypophysitis, drugs, benign adenoma) . **

Reasoning Process :

1 . **Absent Menstruation and Galactorrhea : **

- The patient's absence of menstruation and galactorrhea strongly suggests hyperprolactinemia . Hyperprolactinemia is a well-known cause of both amenorrhea (absent menses) and galactorrhea .

- In a woman of reproductive age , amenorrhea is often associated with oligomenorrhea or an ovulation , which can be due to elevated prolactin levels interfering with normal reproductive function .

2 . **Galactorrhea : **

- Bilateral spontaneous nipple discharge is characteristic of hyperprolactinemia . It occurs because elevated prolactin levels suppress gonadotropin-releasing hormone (GnRH), leading to decreased secretion of follicle-stimulating hormone (FSH) and luteinizing hormone (LH), which in turn reduces estrogen production . This reduced estrogen levels can lead to milk production in the breasts .

3 . **Elevated Prolactin Levels : **

- Prolactin-secreting pituitary adenomas (prolactinomas) are the most common cause of hyperprolactinemia . These adenomas can be solitary or multiple .

- Hypothalamic tumors (e.g., empty sella syndrome) can also cause hyperprolactinemia but are less common in this case given the absence of structural brain abnormalities on MRI .

4 . **Decreased FSH and LH : **

- Elevated prolactin levels directly suppress FSH and LH secretion . This results in decreased gonadotropin levels , leading to decreased menstrual activity , oligomenorrhea , or amenorrhea . The diminished gonadotropins also suppress follicular development , causing normal pituitary morphology on MRI .

5 . **MRI of the Pituitary Gland : **

- The normal MRI result rules out structural pituitary pathology such as tumors , cysts , or masses . This further supports the hypothesis that the cause of hyperprolactinemia is functional , such as a prolactinoma or hypothalamic hypophysitis .

6 . **Clinical Features : **

- The combination of absent menstruation , spontaneous nipple discharge , and normal pituitary morphology strongly suggests a functional cause of hyperprolactinemia , consistent with the pathology of a prolactin-secreting pituitary adenoma .

Conclusion :

The patient's lab results , consistent with elevated prolactin levels , decreased FSH , and decreased LH , are expected in this case due to hyperprolactinemia caused by a prolactin-secreting pituitary adenoma . The absence of structural brain abnormalities on MRI further supports this diagnosis .

</think>

<answer>

Increased prolactin , decreased FSH , decreased LH

</answer>

1.5

1.0

0.5

0.0

-0.5

-1.0

-1.5